2025 Physics Midterm Exam

About the document:

- This exam is from the 2025 Midterm Exam (22 Sep 2025), Section 2 (NANO), and it is not the full exam. I have only included notable problems (No question here is from the in-class exercise set).
- I memorized all these questions personally inside the exam room, please accept that there may be minor errors.
- Although these questions are purely written from memory, I have retained the exact numbers and values used in the actual exam, so treat the math as exact.

Name Student ID No. CR56

2304153 Phys Engs - Midterm Exam (Useful formula)

Gravitational acceleration $g = 9.80$ m/s^2

Calculus: $\frac{d}{dt}(c_1 f(t) \pm c_2 g(t)) = c_1 \frac{d}{dt} f(t) \pm c_2 \frac{d}{dt} g(t)$, $\frac{d}{dt} c\, t^n = c\, n t^{n-1}$, $\frac{d}{dt} e^{ct} = c\, e^{ct}$

$\int c\, t^n dt = c \frac{t^{n+1}}{n+1} + C$ for $n \neq -1$, $\int c \frac{1}{t} dt = c \ln t + C$, $\int e^{ct} dt = \frac{1}{c} e^{ct} + C$

Motion and Laws of Motion: $\vec{v}_{\text{avg}} = \frac{\Delta \vec{r}}{\Delta t}$, $\vec{a}_{\text{avg}} = \frac{\Delta \vec{v}}{\Delta t}$, $\vec{v} = \frac{d}{dt}\vec{r}$, $\vec{a} = \frac{d}{dt}\vec{v}$

$\vec{a} = \text{constant}$; $\Delta \vec{r} = \frac{(\vec{v}_i + \vec{v}_f)}{2} \Delta t$, $\vec{v}_f = \vec{v}_i + \vec{a} t$, $\vec{r}_f = \vec{r}_i + \vec{v}_i t + \frac{1}{2}\vec{a} t^2$, $v_f^2 = v_i^2 + 2\vec{a} \cdot \Delta \vec{r}$

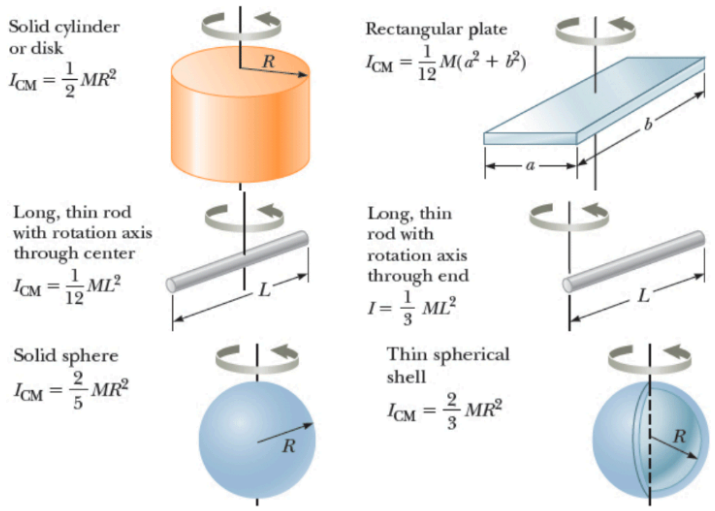
$a_c = \frac{v^2}{r} = \omega^2 r$, $T = \frac{2\pi r}{v}$, $\omega = \frac{2\pi}{T}$, $\vec{a} = \vec{a}_t + \vec{a}_r$, $a = \sqrt{a_t^2 + a_r^2}$

$\Sigma \vec{F} = m\vec{a}$, $f_s \leq \mu_s N$, $f_k = \mu_k N$

Work and energy: $W = \vec{F} \cdot \Delta \vec{r} = F\, \Delta r \cos\theta$, $W = \int \vec{F} \cdot d\vec{r}$, $W_{\text{net}} = \Delta K$, $P = \vec{F} \cdot \vec{v}$

$$K = \frac{1}{2} m v^2, \quad U_g = mgy, \quad U_s = \frac{1}{2} k x^2, \quad W_{\text{CON}} = -\Delta U, \quad F_{\text{CON}} = -\frac{dU}{dx}$$

Moment of Inertia:



Linear momentum and collisions: $\vec{p} = m\vec{v}$, $\vec{I} = \Delta \vec{p} = \int \Sigma \vec{F} dt$

$\vec{r}_{\text{CM}} = \frac{1}{M} \Sigma_i m_i \vec{r}_i = \frac{1}{M} \int \vec{r}\, dm$, $\vec{v}_{\text{CM}} = \frac{1}{M} \Sigma_i m_i \vec{v}_i$, $\vec{F}_{\text{ext}} = M \vec{a}_{CM}$

Rotation of a rigid object about a fixed axis: $\theta = s/r$, $\tau = rF \sin\phi = Fd$

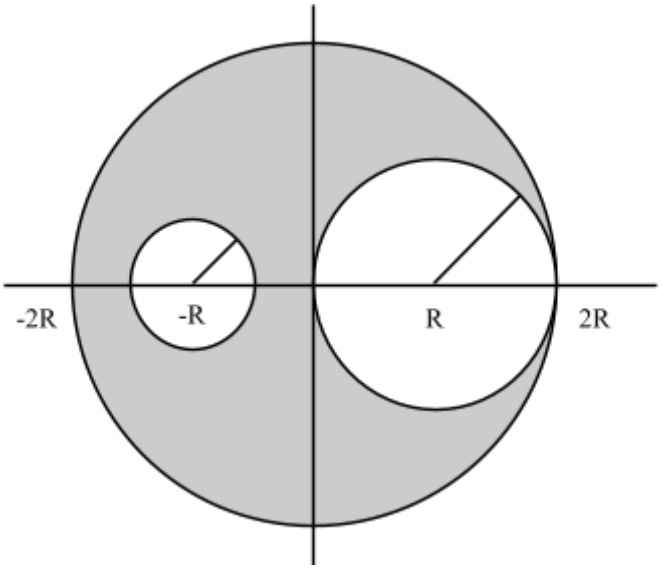
$I = \Sigma_i m_i r_i^2 = \int r^2\, dm$, $I = I_{\text{CM}} + Md^2$, $P = \tau\omega$, $K_{\text{R}} = \frac{1}{2} I \omega^2$, $\Sigma \tau_{\text{ext}} = I\alpha$

$K_{\text{rolling}} = \frac{1}{2} M v_{\text{CM}}^2 + \frac{1}{2} I_{\text{CM}} \omega^2$, $v_{\text{CM}} = \omega R$, $a_{\text{CM}} = \alpha R$

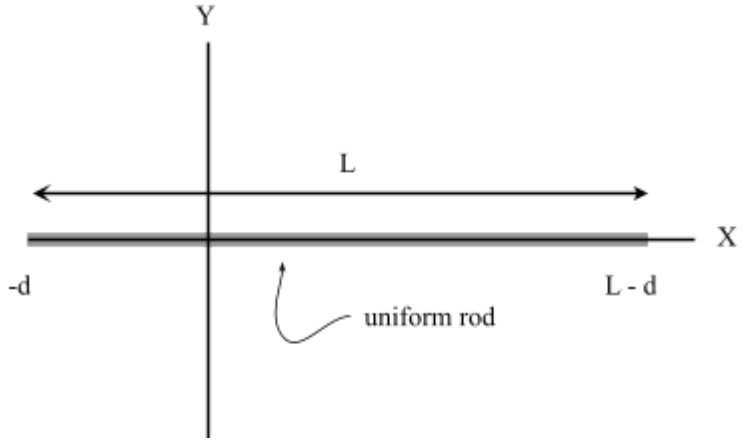
Angular momentum: $|\vec{A} \times \vec{B}| = AB \sin\phi$, $\vec{\tau} = \vec{r} \times \vec{F}$, $\vec{L} = \vec{r} \times \vec{p}$

$\vec{\tau} = I\vec{\alpha} = \frac{d}{dt}\vec{L}$, $\vec{L} = I\vec{\omega}$

Determine the coordinates of the center of mass for the following uniform area mass density system.

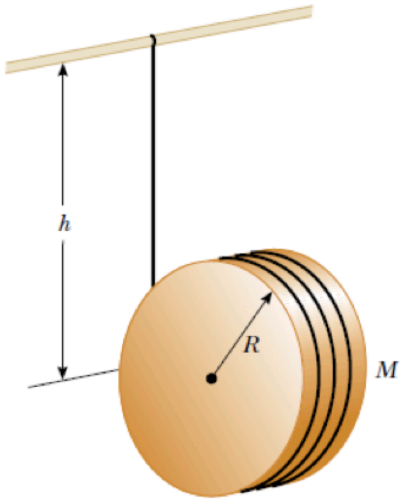


Consider the following uniform mass rod system.



A. Determine the moment of inertia of the given rod around the Y axis (Do not use the Parallel Axis Theorem)
B. Determine A.) with Parallel Axis Theorem
C. In the case of the lowest possible moment of inertia around the Y axis. Evaluate the value of d.
D. Evaluate moment of inertia around the X axis.

A string is wound round a uniform disk of radius R and mass M. The disk is released from rest with the string vertical and its top end tied to a fixed bar.



Proof that

A. The tension in the string is one third of the weight of the disk

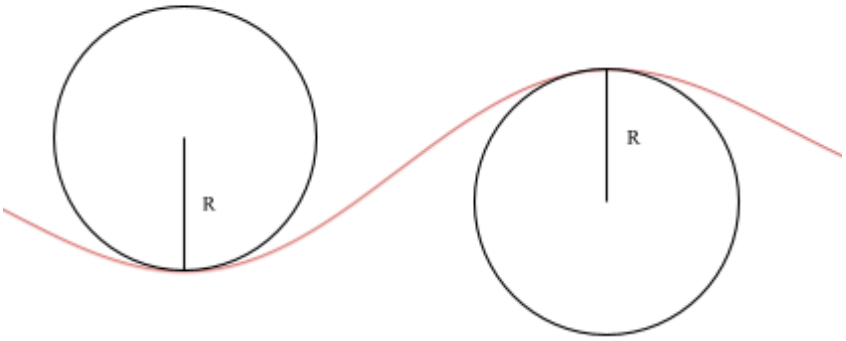
B. The magnitude of the acceleration of the center of mass is $\frac{2g}{3}$

C. The speed of the center of mass is $\sqrt{\frac{4gh}{3}}$ after the disk has descended through distance d (Using Newton's second law in rotational form)

D. Verify your answer to C. using the energy approach.

An object of mass M is released from rest from a height of 2 m. It experiences a linear drag force $f = - Cv$. The object reaches terminal velocity after falling distance of 0.75 m. It then travels the remaining distance to the ground in 2.5 seconds. Determine the drag coefficient C.

A vehicle with mass M = 70 kg travels at a constant speed along a track. The vertical profile of the track consists of a circular hill and a circular valley, both having the same radius of curvature R. At the crest of the hill, the normal force exerted on the vehicle is measured to be 4 N. Determine the normal force exerting on the vehicle at the lowest point of the valley.



A bomber release a payload from height y with a launch speed v and a dive angle θ. It lands at a direct distance d from the release point. Express the vertical motion in terms of these geometric parameters ($y,\ d,\ v,\ \theta$).

(Note: The original problem specified distinct numerical values for $d,\ v,\ y$ and ask for the value of θ; however, as I can't recall them, the general variables are used here.)